

Gene Set Enrichment Analysis: Results Summary

Overview

We tested whether genes identified as auditory phenodeviants by our Bayesian mixture model (133 genes, 139 alleles with $BF \geq 3$) are enriched for other phenotypic categories across the IMPC dataset. The analysis uses Fisher's exact tests with Benjamini-Hochberg FDR correction, applied at every level of the Mammalian Phenotype (MP) ontology hierarchy.

Foreground: 133 unique genes with Bayes Factor ≥ 3 (substantial evidence of hearing loss)

Background: 6,549 unique genes analysed in the Bayesian pipeline (complete ABR data, met inclusion criteria)

Phenotype data source: IMPC genotype-phenotype assertions (67,350 significant calls across all phenotyping procedures, Data Release 22.1)

MP ontology: 14,695 terms, 19,487 parent-child relationships (OBO release 2025-03-19)

Two-pass analysis design

The analysis is structured in two passes to allow both comparability with prior literature and methodologically rigorous assessment of circularity:

Pass 1 — Unfiltered enrichment (Stages 2 and 3): Enrichment is computed using all IMPC genotype-phenotype assertions, replicating the approach used by Vicencio-Jimenez et al. (2026). Enriched terms are labelled by a two-pass classifier (keyword/ontology + procedure-based) as acoustic-dependent, vestibular, or independent — but no assertions are removed. This allows direct comparison with published OR values.

Pass 2 — Assertion-level filtered enrichment (Stage 3b): Individual gene-to-term assertions generated by acoustic-dependent procedures are removed at source before gene-to-MP mappings are rebuilt and enrichment is re-run from scratch. Terms that are partially acoustic-dependent retain their non-acoustic evidence, while terms that are entirely acoustic-derived naturally drop to zero foreground genes and disappear. The change between Pass 1 and Pass 2 quantifies the circularity in each enrichment result.

Stage 2: Top-Level MP Term Enrichment (Unfiltered)

7 of 24 top-level MP categories significantly enriched (FDR < 0.05):

MP Category	OR	Foreground (a)	Background total	Fold enrichment	p_adj
hearing/vestibular/ear	123.29	106	304	17.17	1.15e-121
behavior/neurological	4.64	95	2,340	2.00	3.45e-16
nervous system	4.55	42	633	3.27	8.77e-12
skeleton	2.83	48	1,116	2.12	4.54e-07
reproductive system	2.00	33	943	1.72	4.45e-03
homeostasis/metabolism	1.57	69	2,679	1.27	2.54e-02
immune system	1.60	44	1,558	1.39	3.06e-02

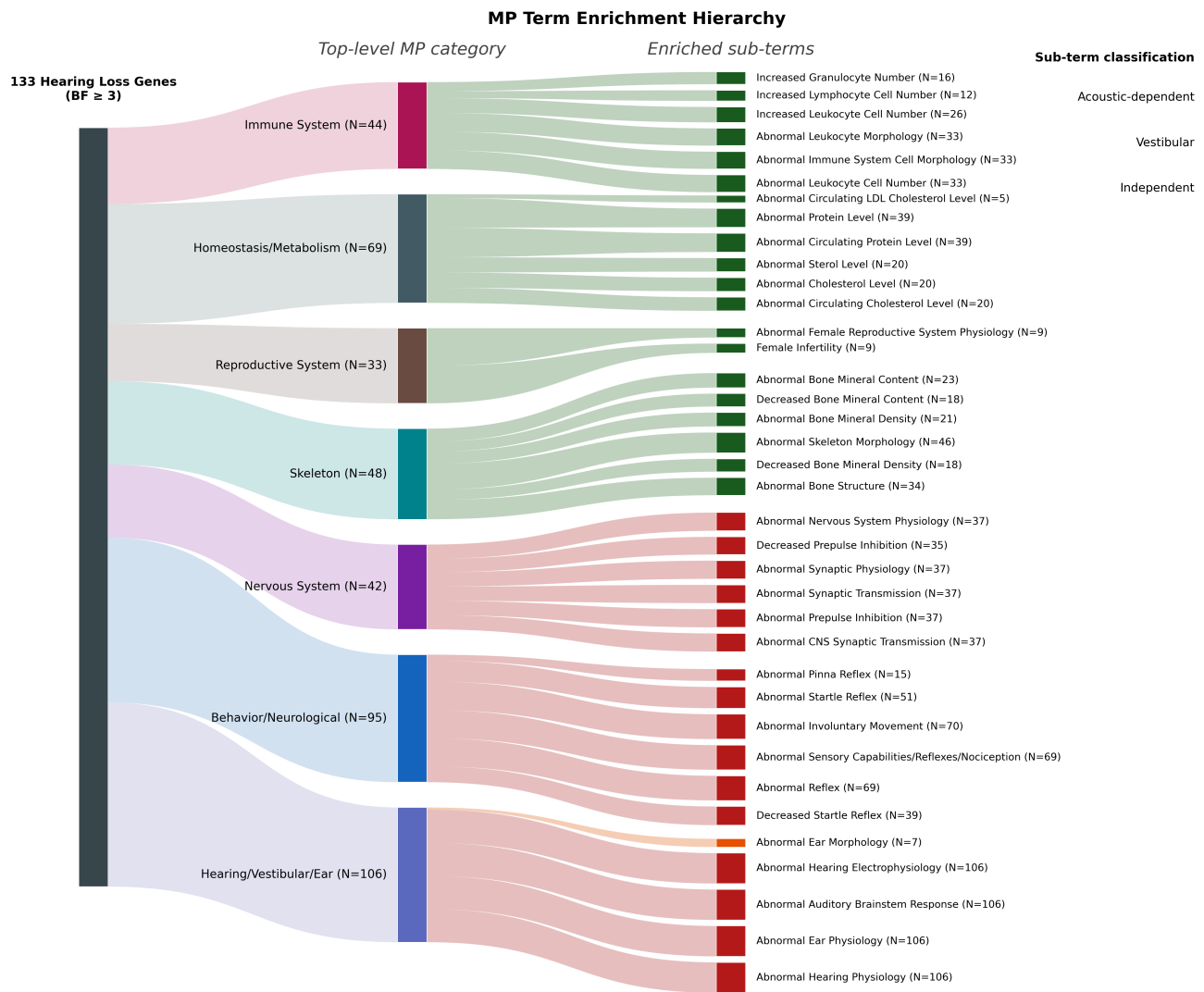


Figure 1. Three-level Sankey diagram showing the flow from 133 hearing loss genes (BF ≥ 3) through significantly enriched top-level MP categories to their enriched sub-terms. Sub-term nodes are coloured by circularity classification: red = acoustic-dependent, orange = vestibular, green = independent.

Comparison with Vicencio-Jimenez et al. (2026)

Vicencio-Jimenez report for their 331 IMPC-annotated hearing loss genes (Data Release 23.0):

- Behavioral: OR=3.13 — we find OR=4.64
- Cognitive: OR=1.98 — we find abnormal cognition OR=2.25 at the sub-branch level
- Emotional: OR=2.11 — we find abnormal emotion/affect OR=2.27 at the sub-branch level

Our analysis additionally identifies skeleton (OR=2.83), reproductive (OR=2.00), homeostasis/metabolism (OR=1.57), and immune (OR=1.60) enrichments not reported by Vicencio-Jimenez.

Stage 3: Hierarchical Enrichment (Unfiltered)

525 terms tested across all descendant branches of the 7 significant top-level categories. 157 terms reached significance (FDR < 0.05).

For visualisation purposes, each enriched term is labelled as acoustic-dependent, vestibular, or independent based on its generating procedures and position in the MP ontology (these labels colour the Sankey diagram and dot plot). The actual circularity assessment is performed by the assertion-level filtering in Stage 3b, which operates independently on the raw data.

Terms generated by acoustic-dependent procedures

The following significant terms are generated entirely or predominantly by procedures that use acoustic stimuli (ABR, acoustic startle/PPI, SHIRPA pinna reflex). These terms are expected to be enriched in hearing loss genes and do not represent independent phenotypic associations.

Term	OR	a	Generating procedures
abnormal hearing physiology	132.23	106	ABR
abnormal ear physiology	132.23	106	ABR
abnormal auditory brainstem response	132.23	106	ABR
abnormal hearing electrophysiology	132.23	106	ABR
decreased startle reflex	33.71	39	Acoustic startle
abnormal startle reflex	10.71	51	Acoustic startle, SHIRPA
abnormal prepulse inhibition	8.17	37	Acoustic startle (PPI)
decreased prepulse inhibition	8.56	35	Acoustic startle (PPI)
abnormal pinna reflex	38.71	15	SHIRPA
absent pinna reflex	38.71	15	SHIRPA
abnormal CNS synaptic transmission	8.17	37	Acoustic startle (PPI) — 37/37 foreground genes
abnormal synaptic transmission	8.17	37	Acoustic startle (PPI) — 37/37 foreground genes

Term	OR	a	Generating procedures
abnormal synaptic physiology	8.17	37	Acoustic startle (PPI) — 37/37 foreground genes
abnormal nervous system physiology	7.80	37	Acoustic startle (PPI) — 37/37 foreground genes
abnormal reflex	11.39	69	Acoustic startle + SHIRPA — 66/69 foreground genes
abnormal sensory capabilities/reflexes/nociception	11.19	69	Acoustic startle + SHIRPA — 66/69 foreground genes
abnormal involuntary movement	10.61	70	Acoustic startle + SHIRPA — 63/70 foreground genes
limb grasping	3.48	8	SHIRPA — 7/8 foreground genes

Note: terms such as "abnormal synaptic transmission" and "abnormal nervous system physiology" have non-acoustic names but are generated almost entirely by acoustic startle/PPI procedures. Without procedure-level scrutiny, these would be misreported as independent neurological enrichments.

Vestibular terms (inner ear co-morbidity)

Term	OR	a	p_adj
abnormal ear morphology	25.40	7	1.12e-07
head bobbing	37.84	6	1.03e-06
trunk curl	22.22	7	1.22e-06

Other enrichment

Behavioral / Neurological

Term	OR	a	p_adj	Procedure type
hyperactivity	2.77	31	2.97e-05	Open field

Term	OR	a	p_adj	Procedure type
increased vertical activity	4.65	12	1.43e-04	Open field
decreased anxiety-related response	3.56	15	2.68e-04	Open field / Light-dark
abnormal anxiety-related response	3.32	17	2.32e-04	Open field / Light-dark
abnormal gait	4.39	11	3.65e-04	SHIRPA / Grip strength
abnormal associative learning	7.17	6	1.05e-03	Fear conditioning
abnormal cognition	2.25	19	5.52e-03	Fear conditioning / Open field
abnormal contextual conditioning	7.07	4	8.06e-03	Fear conditioning
decreased grip strength	2.51	13	8.31e-03	Grip strength
decreased locomotor activity	2.34	15	8.31e-03	Open field
tremors	3.59	5	2.88e-02	SHIRPA
increased thermal nociceptive threshold	97.94	2	3.13e-03	Hot plate / Hargreaves

Skeleton

Term	OR	a	p_adj
decreased bone mineral density	4.72	18	8.85e-06
abnormal bone mineral density	4.03	21	9.11e-06
abnormal bone structure	3.05	34	8.85e-06

Term	OR	a	p_adj
abnormal radius morphology	49.33	3	9.79e-04
abnormal ulna morphology	49.33	3	9.79e-04
abnormal humerus morphology	36.99	3	1.50e-03

Homeostasis / Metabolism

Term	OR	a	p_adj
abnormal circulating cholesterol level	2.85	20	7.38e-03
abnormal circulating LDL cholesterol level	9.99	5	7.93e-03
increased circulating alkaline phosphatase	2.68	19	8.14e-03
abnormal lipid level	2.33	24	8.40e-03
abnormal glucose homeostasis	1.85	28	2.64e-02

Reproductive

Term	OR	a	p_adj
female infertility	5.22	9	2.68e-03

Immune

Term	OR	a	p_adj
abnormal leukocyte cell number	2.28	33	6.11e-03
increased leukocyte cell number	2.30	26	1.55e-02
decreased T cell number	4.62	6	3.78e-02



Figure 2. Dot plot of enriched MP sub-terms (FDR < 0.05), ordered by classification: acoustic-dependent (top, red), vestibular (middle, orange), independent (bottom, green). Dot size is proportional to the number of foreground genes. Fold enrichment indicates the ratio of the observed to expected rate of the term in foreground genes.

Stage 3b: Assertion-Level Acoustic Filtering

Rather than classifying whole MP terms and applying an arbitrary threshold, we filter at the individual assertion level:

- **Fully acoustic procedures** (ACS, ABR, ESLIM_011): all 2,774 assertions removed
- **Mixed procedures** (CSD/SHIRPA): 253 assertions with acoustic-specific MP terms (pinna reflex, startle) removed; 3,728 non-acoustic assertions (gait, coat morphology, posture) retained
- **All other procedures**: all 60,595 assertions retained
- **Total removed**: 3,027 of 67,350 assertions (4.5%)
- **Remaining**: 64,323 assertions

Gene-to-MP mappings are rebuilt from scratch on the filtered data and enrichment re-run.

Top-level: full vs assertion-filtered comparison

MP Category	Full OR	Full a	Filtered OR	Filtered a	Full sig?	Filtered sig?
nervous system	4.55	42	1.02	7	YES	no
behavior/neurological	4.64	95	2.49	72	YES	YES
hearing/vestibular/ear	123.29	106	25.40	7	YES	YES
skeleton	2.83	48	2.83	48	YES	YES
reproductive system	2.00	33	2.00	33	YES	YES
homeostasis/metabolism	1.57	69	1.57	69	YES	YES
immune system	1.60	44	1.60	44	YES	YES

The nervous system enrichment collapses entirely (OR 4.55 → 1.02, a drops from 42 to 7). The behavioral enrichment is partially reduced (OR 4.64 → 2.49, a drops from 95 to 72 as 23 genes lose their acoustic-derived behavioral assertions) but remains significant. Skeleton, reproductive, metabolic, and immune enrichments are completely unaffected — none of their evidence comes from acoustic procedures.

Filtered hierarchical enrichment

143 significant terms survive assertion-level filtering (down from 157). The 14 terms that drop out are exactly the acoustic-dependent terms identified by the classifier: all ABR/hearing terms, startle/PPI terms, pinna reflex, and their procedure-driven parent terms (synaptic transmission, nervous system physiology, abnormal reflex).

Key terms that survive with reduced gene counts:

Term	Full a	Filtered a	Filtered OR	Source
abnormal motor capabilities/coordination/movement	87	58	2.41	Open field, grip strength, SHIRPA (non-acoustic items)
abnormal behavior	95	72	2.49	Open field, light-dark, fear conditioning
abnormal involuntary movement	70	—	dropped	Was 90% acoustic
abnormal reflex	69	—	dropped	Was 96% acoustic

All skeleton, metabolic, immune, and reproductive terms retain identical gene counts — they had zero acoustic contributions.

Stage 4: Centre-Matched Sensitivity Analysis

Centre-matched results (only counting MP assertions from the same centre as the gene's ABR data) are consistent with the unmatched analysis:

MP Category	Unmatched OR	Centre-matched OR	Centre-matched sig?
hearing/vestibular/ear	123.29	123.94	YES
behavior/neurological	4.64	4.57	YES
nervous system	4.55	4.50	YES
skeleton	2.83	2.71	YES

MP Category	Unmatched OR	Centre-matched OR	Centre-matched sig?
reproductive system	2.00	2.01	YES
homeostasis/metabolism	1.57	1.62	YES
immune system	1.60	1.64	YES
adipose tissue	1.88 (NS)	1.92	YES (p_adj=0.043)

Centre matching slightly strengthens some effects and brings adipose tissue to significance. The enrichment patterns are not driven by cross-centre confounding. The high match rate (47,669 matched vs 1,767 unmatched assertions, 96.4%) reflects the IMPC's standardised pipeline where most phenotyping occurs at the same centre.